Medicaid Expansion and Spending by Payer

Working results after correcting the treatment coding

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31 August 2026 · DEX 2.0, 2010–2018

Seven states were coded as 2014 expanders that had not expanded by 2019

34

states coded as expanding in 2014

27

states that actually did

ID, UT, NE, MO, OK, SD and NC expanded in 2020 or later. All were in the treated group.

Cohorts are now assigned by first full calendar year of exposure

Cohort	n	States
2014	25	AZ AR CA CO CT DE DC HI IL IA KY MD MA MN NV NJ NM NY ND OH OR RI VT WA WV
2015	3	MI (Apr 2014), NH (Aug 2014), PA (Jan 2015)
2016	3	IN (Feb 2015), AK (Sep 2015), MT (Jan 2016)
2017	1	LA (Jul 2016)
Never	19	AL FL GA ID KS ME MS MO NE NC OK SC SD TN TX UT VA WI WY

Panel truncated at 2018. ME and VA expanded in 2019, so they serve as controls. The control group grew from 10 states to 19.

The doubly robust estimator cannot support five covariates at 51 states

Covariates in propensity score	Doubly robust	IPW
Poverty only	75.9 (12.8)	76.0 (12.6)
Prior uninsurance only	77.5 (17.3)	76.5 (14.4)
Uninsurance + income	73.0 (14.5)	74.2 (12.8)
Poverty + income	fails	fails
Three covariates	fails	fails
All five	fails	fails

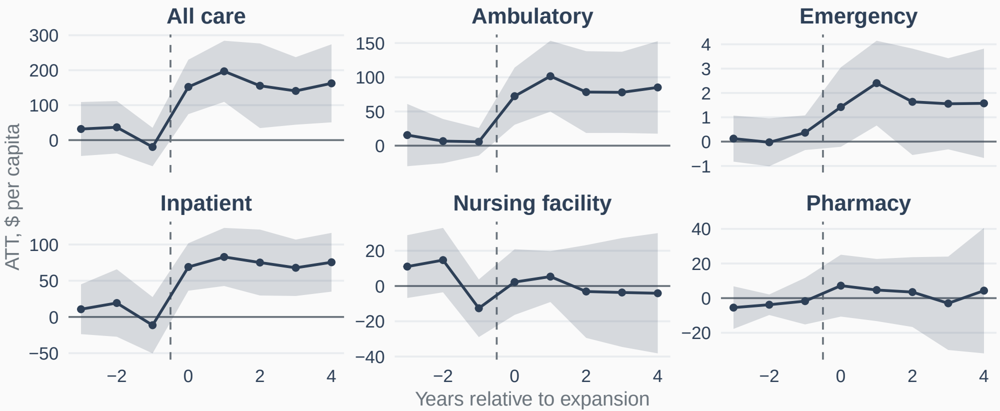
Medicaid inpatient spending per capita. "Fails" = fitted probabilities of 0 or 1, leaving group-time cells unestimated. Not a thin-cohort artifact: it fails on all 8 cells of the 2014 cohort alone (25 treated vs 19 control).

Main spec: outcome regression, five covariates. **Sensitivity:** doubly robust, uninsurance + income.

Results

2014 cohort vs. never-expanded states, outcome regression

Medicaid per-capita spending rises in ambulatory and inpatient care



Callaway–Sant'Anna group-time ATTs, dynamic aggregation. Band is 95% CI; dashed line is the last pre-expansion year.

Medicaid per-capita estimates, 2014 cohort

Type of care	ATT ($)	SE	t	Pre-test p
All care	161.6	48.4	3.34	0.59
Ambulatory	83.1	27.2	3.06	0.81
Inpatient	74.1	17.9	4.14	0.72
Emergency	1.7	1.0	1.77	0.82
Pharmacy	3.4	12.1	0.28	0.49
Nursing facility	-0.7	12.2	-0.05	0.26

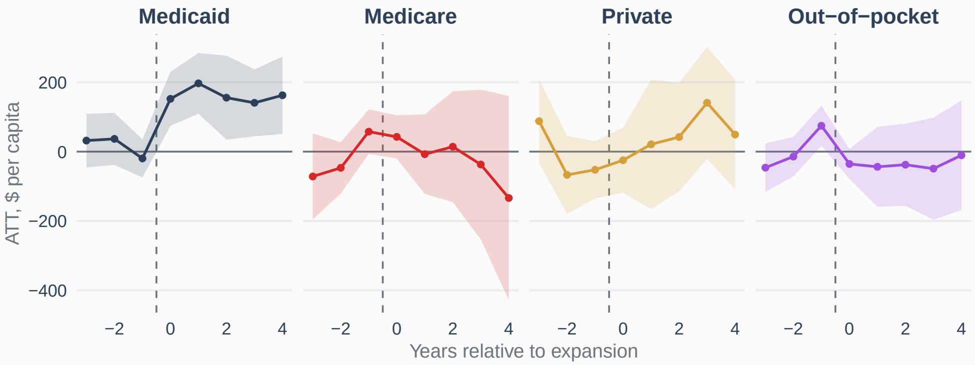
Pre-test is the joint Wald test on pre-expansion group-time cells; higher p means less evidence against parallel trends. On the full staggered sample the same estimates are 169.5, 90.3, 71.8, 2.0, 8.0 and -2.6.

Medicaid spending per beneficiary is statistically indistinguishable from zero

Type of care	ATT ($)	SE	t	Pre-test p
All care	-299.2	260.3	-1.15	0.37
Ambulatory	-70.5	131.9	-0.53	0.62
Inpatient	68.2	96.3	0.71	0.19
Emergency	-4.0	4.4	-0.90	0.52
Pharmacy	-134.9	97.9	-1.38	0.18
Nursing facility	-158.1	131.3	-1.20	0.51

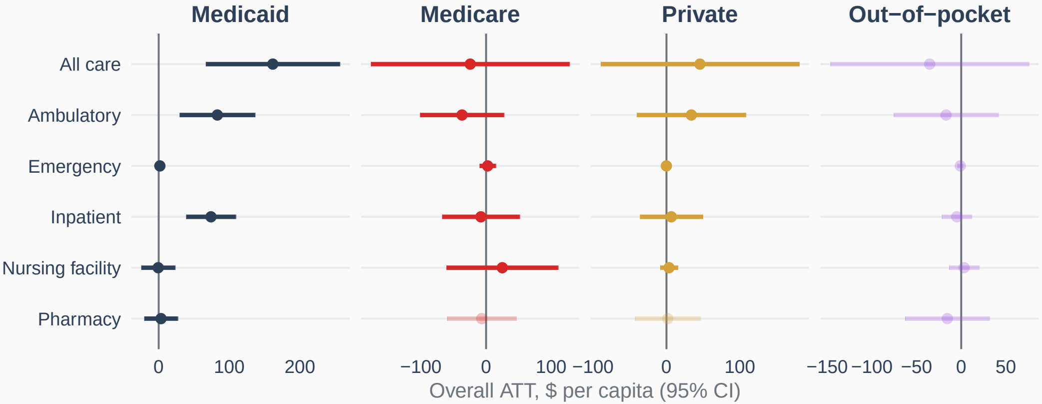
Same models as the previous slide, per Medicaid beneficiary rather than per state resident.

Across all care, only the Medicaid series moves at expansion



Overall ATTs, $ per capita: Medicaid 161.6 (SE 48.4), Medicare -24.3 (77.7), private 45.6 (68.9), out-of-pocket -35.3 (56.7). The out-of-pocket model fails the pre-trend test at $p = 0.0001$.

Per-capita estimates by payer and setting, with pre-test status

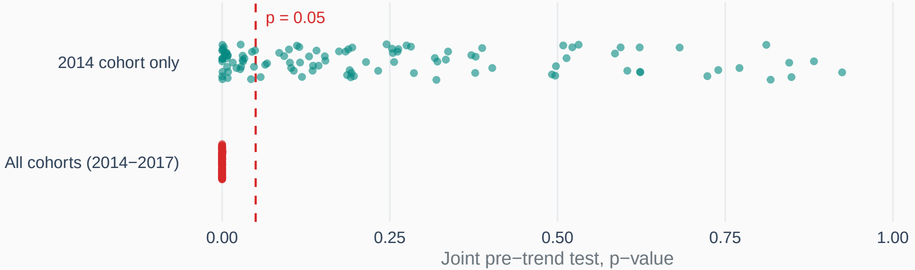


Faded points fail the joint pre-trend test at $p < 0.05$. Horizontal bars are 95% CI. Note the free horizontal scale across panels.

Diagnostics

What the pre-trend test can and cannot tell us

The pre-trend test rejects every model on the staggered sample



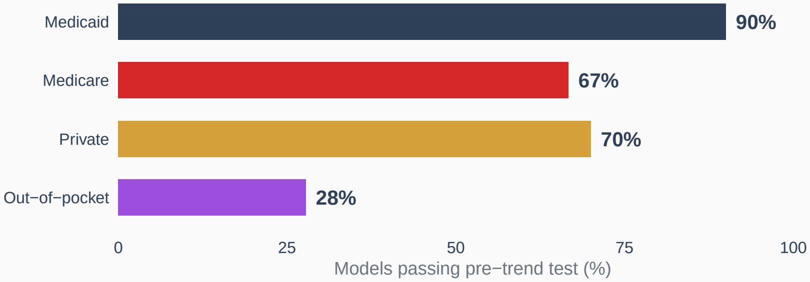
Each point is one payer–setting–outcome combination ($n = 108$). 0 of 108 pass on the staggered sample; 73 of 108 pass on the 2014 cohort.

The pre-test is oversized when treated cohorts are small

Fake cohort size	Pre-test rejects	ATT falsely sig.
1 state	88.5%	40.0%
2 states	76.0%	—
3 states	65.0%	11.5%
6 states	33.5%	12.5%
10 states	36.5%	10.5%

Placebo: never-expanded states only, k of them labelled as a fake cohort. No treatment exists, so a correctly sized test rejects 5% of the time. The real cohorts are 25, 3, 3 and 1 states. 200 replications each; `R/pretest_placebo.R`.

Pre-test pass rates differ sharply by payer



2014 cohort, main specification, across all types of care and all five outcomes ($n = 30$ per payer; 18 for out-of-pocket, which has no per-beneficiary denominator). All six out-of-pocket spending-per-capita models fail, with p from 0.0001 to 0.049.

The two specifications agree on sign in 14 of 16 significant estimates

0.81

correlation among estimates
significant in the main spec

0.62

correlation across
all 108 estimates

Main: outcome regression, five covariates. Sensitivity: doubly robust, uninsurance + income. Both sign flips are Medicare unit-price outcomes (inpatient, nursing facility), neither significant in the sensitivity spec.

Open

Choices not yet made

Four decisions remain

OPEN **Primary sample.** The pre-trend test is only interpretable on the 2014 cohort. Making it primary trades 7 treated states for a testable assumption.

OPEN **Out-of-pocket.** Every per-capita model fails the pre-test. Report as unidentified, drop, or change design.

OPEN **Louisiana.** Alone in the 2017 cohort with two post-periods. Currently retained.

OPEN **Waiver states.** The `restricted` sensitivity is coded but not yet run across the matrix.

Everything here is

432 models, all converged

2010–2018, 51 states, four payers, six settings, five outcomes

Estimates: `results/overall_att.csv` · Pre-tests: `results/pretrend_tests.csv`

Pipeline: `R/prep_data.R` → `expansion_CSA.R` → `slides/make_figures.R`